

Tsai-Shien Chen

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Education

University of California, Merced

Merced, CA

PH.D. CANDIDATE IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

Sep. 2022 - Now

- Advisor: Prof. Mukesh Singhal

National Taiwan University

Taipei, Taiwan

M.S. IN ELECTRONICS ENGINEERING

Sep. 2019 - March 2022

- Advisor: Prof. Shao-Yi Chien

B.S. IN ELECTRICAL ENGINEERING

Sep. 2015 - June 2019

- GPA: 4.23 / 4.30; Valedictorian

Research Interests

- Image, Video, World Generative Models
- Video Understanding, Generation, and Application
- Controllable and Personalized Content Creation

Honors & Awards

2026	CVPR Outstanding Reviewers (top 5% of reviewers)
2026	CVPR Doctoral Consortium
2025	Google PhD Fellowship
2024	Graduate Student Opportunity Program Fellowship (one year of full tuition and stipend), UC Merced
2023	Best Master Thesis of the Year Award , Graduate Institute of Electronics Engineering, NTU
2022	Honorary Member (top 2% in college) , Phi Tau Phi Scholastic Honor Society
2019	Valedictorian , Department of Electrical Engineering, NTU
2015-2019	4-time Presidential Award (top 5% in department) , NTU

Work Experiences

Adobe Research

San Francisco, CA

RESEARCH INTERN

March 2026 - Now

- Mentors: Richard Zhang, Eli Shechtman
- Real-time autoregressive video streaming model

Snap Inc.

Santa Monica, CA

RESEARCH INTERN

May 2023 - March 2026

- Mentors: Aliaksandr Siarohin, Sergey Tulyakov, Jun-Yan Zhu, Kfir Aberman
- Large-scale video dataset and video captioning, generation, and personalization models

Publications

Conference Proceedings

1. **CineOrchestra: Unified Entity-Centric Conditioning for Cinematic Video Generation**
Sharath Girish*, **Tsai-Shien Chen***, Zhikang Dong, Mukesh Singhal, Hao Chen, Sergey Tulyakov, Aliaksandr Siarohin
NeurIPS 2026
2. **CanvasComposer: Personalized Group Photo Generation via a Multi-Reference Canvas**
Gordon Guocheng Qian, Ruihang Zhang, **Tsai-Shien Chen**, Yusuf Dalva, Anujraaj Argo Goyal, Willi Menapace, Ivan Skorokhodov, Meng Dong, Arpit Sahni, Daniil Ostashev, Ju Hu, Mukesh Singhal, Sergey Tulyakov, Kuan-Chieh Jackson Wang
SIGGRAPH Asia 2026
3. **Canvas-to-Image: Compositional Image Generation with Multimodal Controls**
Yusuf Dalva, Gordon Guocheng Qian, Maya Goldenberg, **Tsai-Shien Chen**, Kfir Aberman, Sergey Tulyakov, Pinar Yanardag, Kuan-Chieh Jackson Wang
SIGGRAPH 2026

4. **Omni-Attribute: Open-vocabulary Attribute Encoder for Visual Concept Personalization**
Tsai-Shien Chen, Aliaksandr Siarohin, Gordon Guocheng Qian, Kuan-Chieh Jackson Wang, Egor Nemchinov, Moayed Haji-Ali, Riza Alp Guler, Willi Menapace, Ivan Skorokhodov, Anil Kag, Jun-Yan Zhu, Sergey Tulyakov
CVPR 2026
5. **AlcheMinT: Fine-grained Temporal Control for Multi-Reference Consistent Video Generation**
 Sharath Girish, Viacheslav Ivanov, **Tsai-Shien Chen**, Hao Chen, Aliaksandr Siarohin, Sergey Tulyakov
CVPR 2026
6. **Multi-subject Open-set Personalization in Video Generation**
Tsai-Shien Chen, Aliaksandr Siarohin, Willi Menapace, Yuwei Fang, Kwot Sin Lee, Ivan Skorokhodov, Kfir Aberman, Jun-Yan Zhu, Ming-Hsuan Yang, Sergey Tulyakov
CVPR 2025
7. **VIMI: Grounding Video Generation through Multi-modal Instruction**
 Yuwei Fang, Willi Menapace, Aliaksandr Siarohin, **Tsai-Shien Chen**, Kuan-Chieh Jackson Wang, Ivan Skorokhodov, Graham Neubig, Sergey Tulyakov
EMNLP 2024
8. **Panda-70M: Captioning 70M Videos with Multiple Cross-Modality Teachers**
Tsai-Shien Chen, Aliaksandr Siarohin, Willi Menapace, Ekaterina Deyneka, Hsiang-wei Chao, Byung Eun Jeon, Yuwei Fang, Hsin-Ying Lee, Jian Ren, Ming-Hsuan Yang, Sergey Tulyakov
CVPR 2024
9. **Snap Video: Scaled Spatiotemporal Transformers for Text-to-Video Synthesis**
 Willi Menapace, Aliaksandr Siarohin, Ivan Skorokhodov, Ekaterina Deyneka, **Tsai-Shien Chen**, Anil Kag, Yuwei Fang, Aleksei Stoliar, Elisa Ricci, Jian Ren, Sergey Tulyakov
CVPR 2024 (Highlight Presentation)
10. **Incremental False Negative Detection for Contrastive Learning**
Tsai-Shien Chen, Wei-Chih Hung, Hung-Yu Tseng, Shao-Yi Chien, Ming-Hsuan Yang
ICLR 2022
11. **Hard Samples Rectification for Unsupervised Cross-domain Person Re-identification**
 Chih-Ting Liu, Man-Yu Lee, **Tsai-Shien Chen**, Shao-Yi Chien
ICIP 2021
12. **Tracklet-Refined Multi-Camera Tracking Based on Balanced Cross-Domain Re-Identification for Vehicles**
 Kai-Siang Yang, Yu-Kai Chen, **Tsai-Shien Chen**, Chih-Ting Liu, Shao-Yi Chien
CVPR Workshops 2021
13. **Orientation-aware Vehicle Re-identification with Semantics-guided Part Attention Network**
Tsai-Shien Chen, Chih-Ting Liu, Chih-Wei Wu, Shao-Yi Chien
ECCV 2020 (Oral Presentation)
14. **Viewpoint-Aware Channel-Wise Attentive Network for Vehicle Re-Identification**
Tsai-Shien Chen, Man-Yu Lee, Chih-Ting Liu, Shao-Yi Chien
CVPR Workshops 2020
15. **Supervised Joint Domain Learning for Vehicle Re-Identification**
 Chih-Ting Liu, Man-Yu Lee, Chih-Wei Wu, Bo-Ying Chen, **Tsai-Shien Chen**, Yao-Ting Hsu, Shao-Yi Chien
CVPR Workshops 2019

Patents

1. **Captioning Videos with Multiple Cross-Modality Teachers**
Tsai-Shien Chen, Yuwei Fang, Hsin-Ying Lee, Jian Ren, Aliaksandr Siarohin, Sergey Tulyakov
US12489956B2
2. **Scaled Spatiotemporal Transformers for Text-to-Video Synthesis**
Tsai-Shien Chen, Yuwei Fang, Anil Kag, Willi Menapace, Jian Ren, Aliaksandr Siarohin, Ivan Skorokhodov, Sergey Tulyakov
US20250245897A1

Professional Activities

- Invited Talks
 - **UC Merced** (Sep 2026): “Toward Controllable Video Generation: From Web-Scale Data to Efficient Cinematic Control”
 - **ByteDance** (July 2025): “From Data to Director: Towards Controllable and Personalized Movie Generation”
 - **AI Time** (Dec 2025): “From Personalization Models towards Movie Generator”
 - **Anuttacon** (July 2025): “Learning Video Generation and Personalization from Large-Scale Datasets”
- Conf. Reviewer
 - **CVPR**’22,’23,’24,’25,’26; **ICCV**’21,’23,’25; **ECCV**’22,’24,’26; **WACV**’26; **ACCV**’24
 - **NeurIPS**’23,’24,’25,’26; **ICLR**’24,’25,’26,’27; **ICML**’24,’25,’26; **AAAI**: 2026
 - **SIGGRAPH**’26; **SIGGRAPH Asia**’26; **ACMMM**’25,’26
- Journal Reviewer
 - IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**)
 - IEEE Transactions on Image Processing (**TIP**)
 - International Journal of Computer Vision (**IJCV**)
- Teaching Assistant
 - UC Merced: CSE-185 **Introduction to Computer Vision** (Spring 2023)
 - UC Merced: CSE-024 **Advanced Programming** (Fall 2022)
 - NTU: EEE-5053 **Computer Vision** (Spring 2021)